



INTERNATIONAL
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The exciting world of IoT

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Bio (Periklis Chatzimisios)

- ❑ *Professor, Department of Information and Electronic Engineering, International Hellenic University (IHU), Greece*
- ❑ *Research Professor, Department of Electrical and Computer Engineering, University of New Mexico, USA*
- ❑ **Communications & Internet of Things (CIoT) Research Group**
- ❑ **Visiting Fellow at Bournemouth University, Visiting Professor at University Pierre and Marie Curie (UPMC LIP6), University of Toronto and Massachusetts Institute of Technology (MIT)**
- **Member of the IEEE Communication Society (ComSoc) Standards Development Board, IEEE ComSoc Communications Standardization Programs Development Board, IEEE ComSoc Education & Training Board**
- **Vice-Chair for WG 1 (Use cases, KPIs and future market and business scenarios), Chair of the Working Item 213 (Multimodal sensing, computing, communication and control for 6G remote operation) and Board Member of one6G Association**
- **Editorial Member/Chair/Organizing Committee of several journals + conferences**



Internet of Things (IoT)

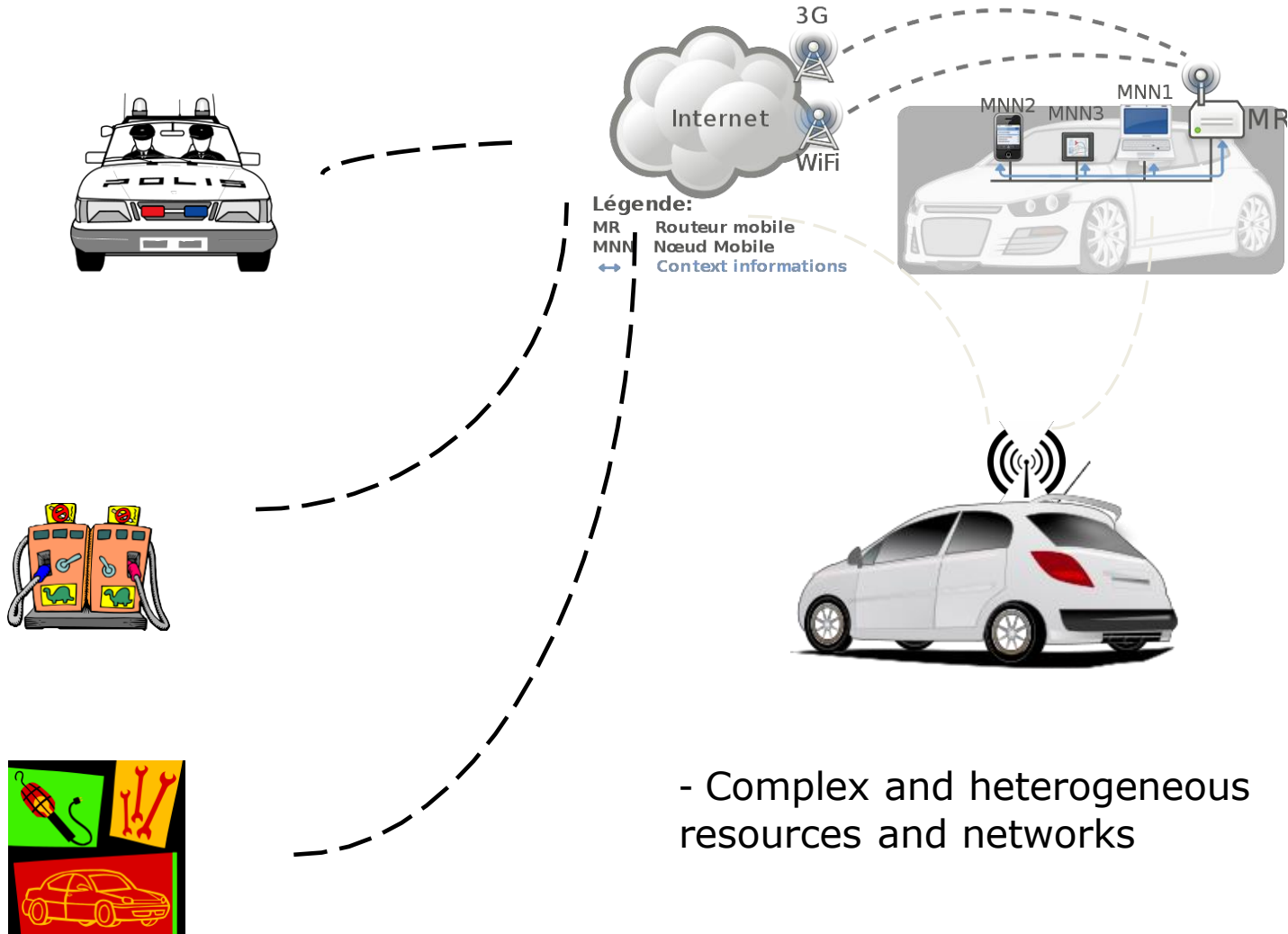


Internet of Everything

Internet of Things

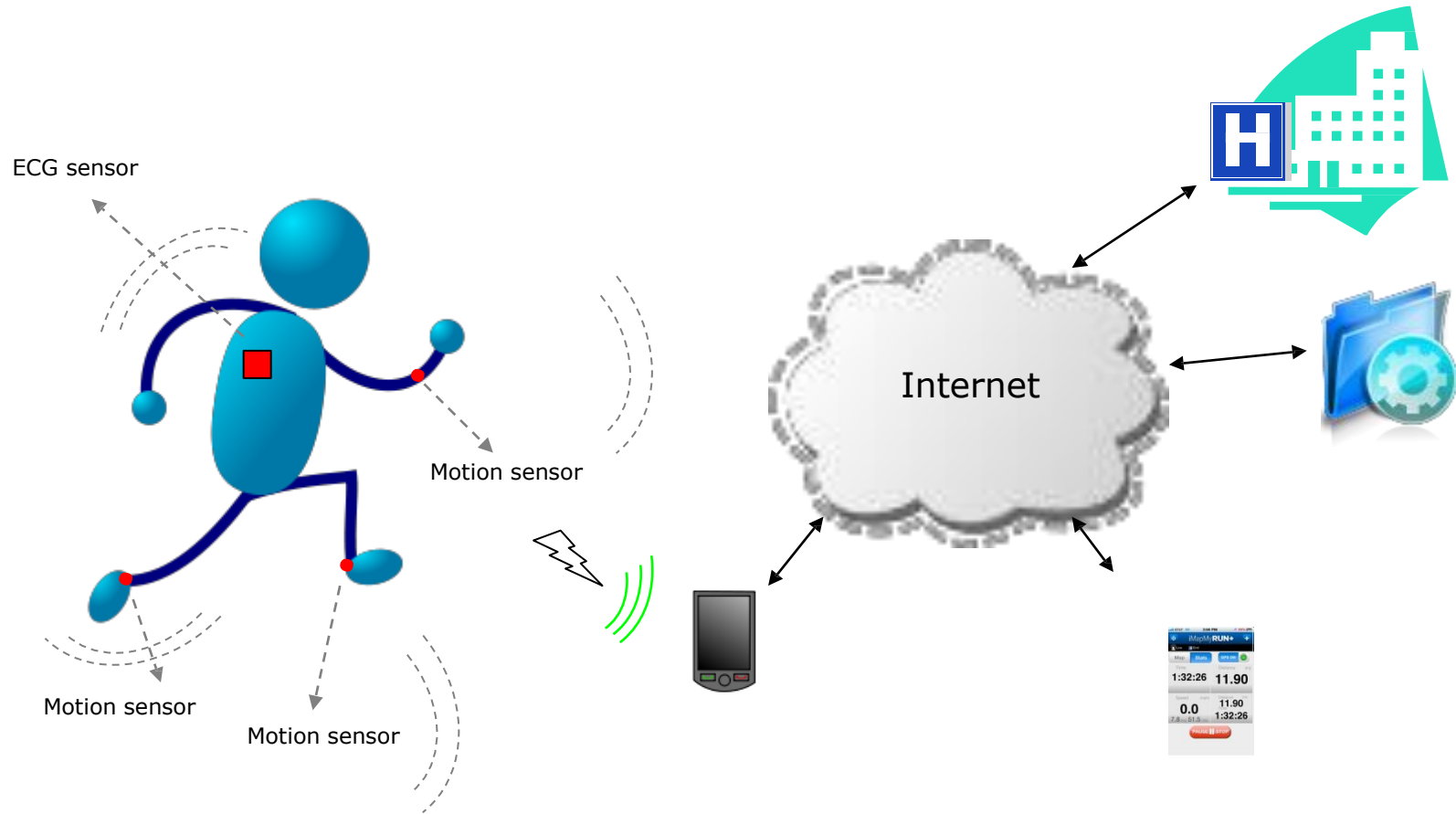
Internet of Things (IoT) is everywhere

Things Connecting to Things



Internet of Things (IoT) is everywhere

People Connecting to Things

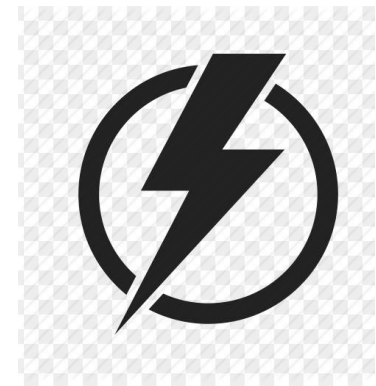


Industrial Internet of Things (IIoT)

Industrial Internet of Things: When the IoT is extended for industrial applications such as manufacturing and supply chain management.

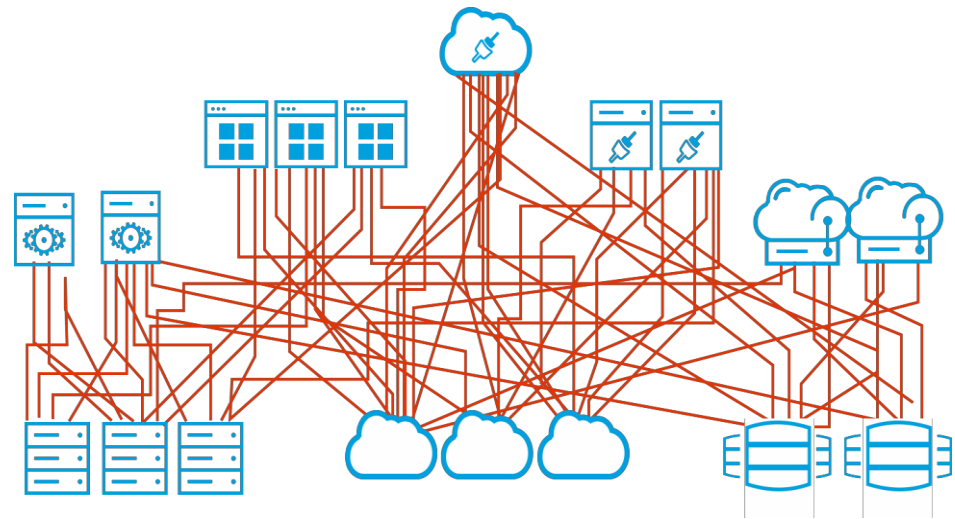
IIoT is an industrial application of IoT (a network connecting industrial machinery and devices) in areas such as oil and gas, transportation, mining and metals, mining, aviation, logistics, etc.

- **Predictive Maintenance**
- **Adaptive Assembly Lines / Customized Products**

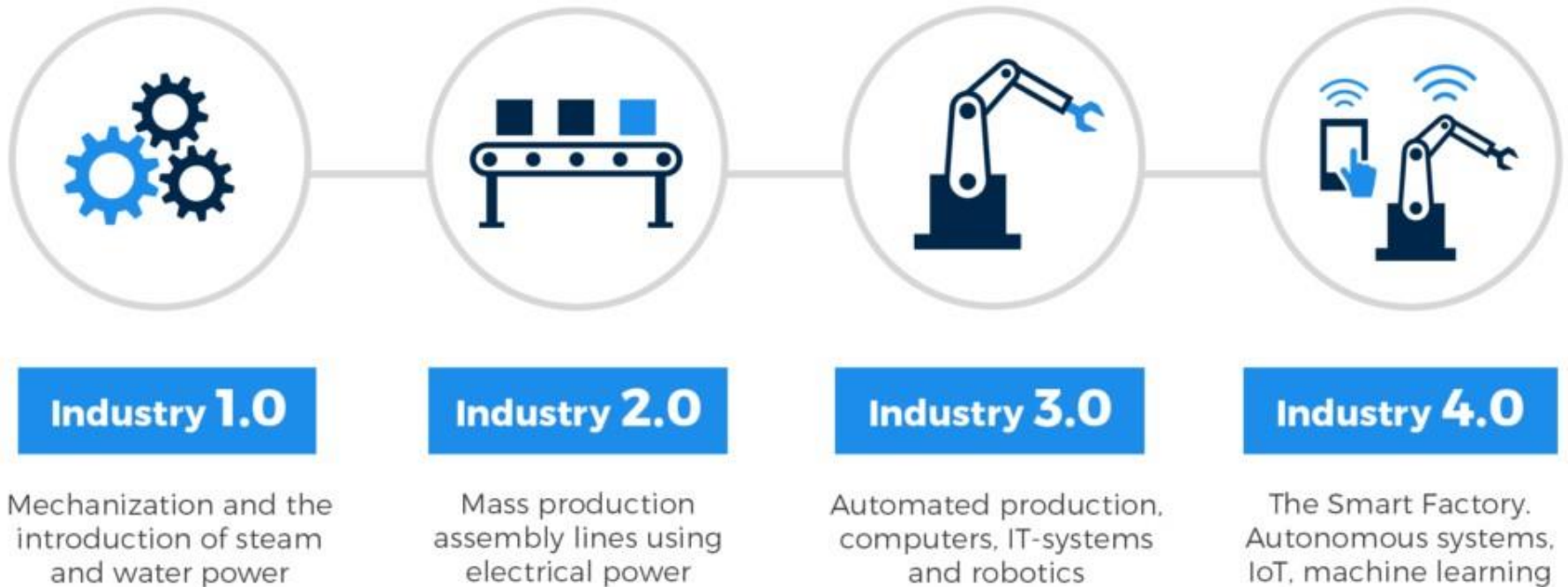


Characteristics of IIoT

- ❑ Machine-to-Machine communication
- ❑ Coexistence & Interoperability
- ❑ Big Data



The Four Industrial Revolutions



Internet of Things (IoT) is everywhere

Sensor devices are becoming widely available

- Programmable devices
- Gadgets/tools



IoT Verticals

Smart
Wearables



Smart
Home



Smart
City



Smart
Agriculture



Connected
Car



Health
Care



Industry
Automation



Smart
Energy



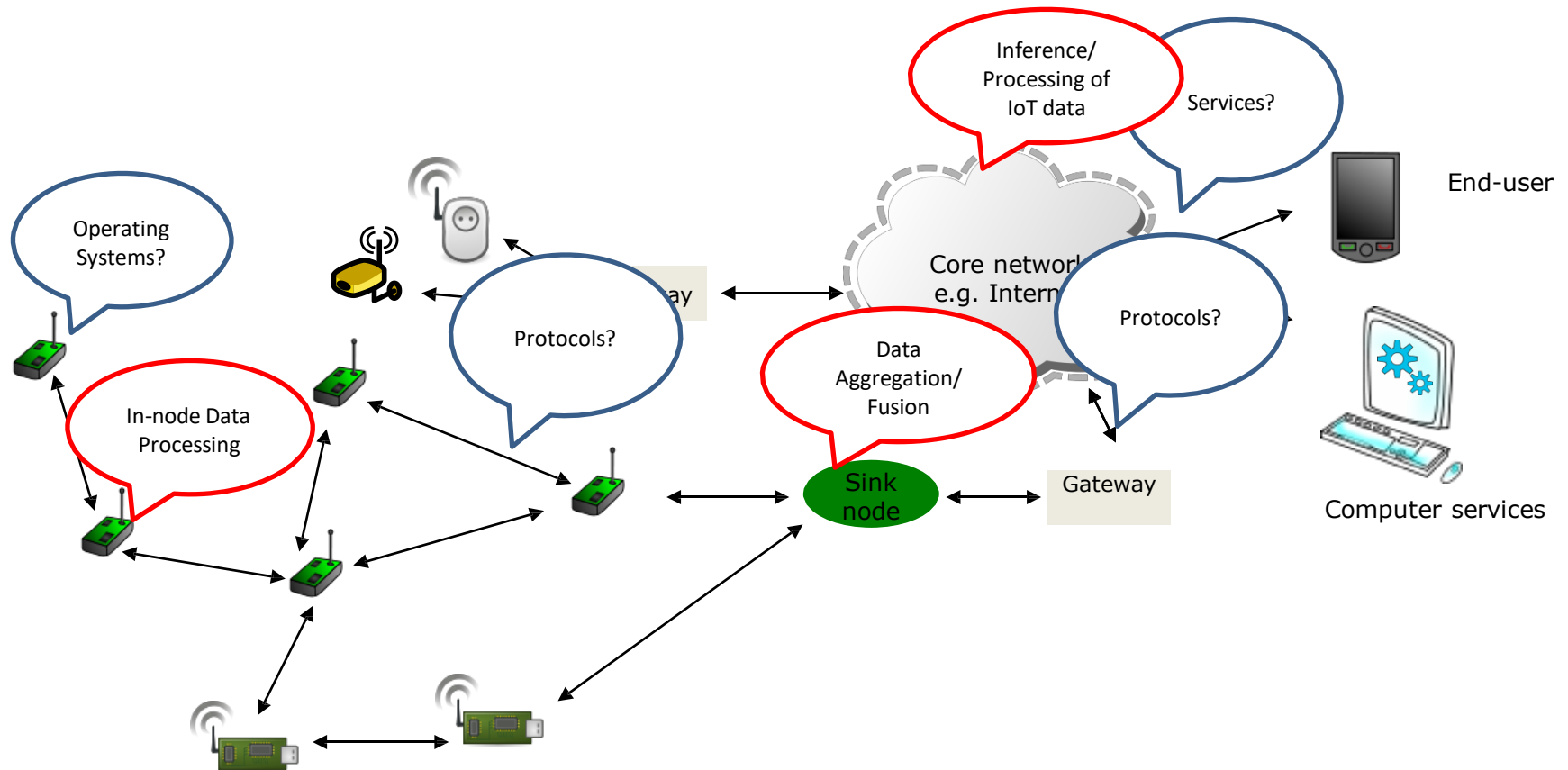
Source: Keysight Technologies



Smart City applications

- **Smart parking:** Monitoring of parking spaces availability in the city.
- **Structural Health:** Monitoring of vibrations and material conditions in buildings, bridges and historical monuments.
- **Traffic Congestion:** Monitoring of vehicles and pedestrian levels to optimize driving and walking routes.
- **Smart lighting:** Intelligent and weather adaptive lighting in street lights.
- **Waste management:** Detection of rubbish levels in containers to optimize the trash collection routes.
- **Smart roads:** Intelligent Highways with warning messages and diversions according to climate conditions and unexpected events like accidents or traffic jams.

Wireless Sensor (and Actuator) Networks



- The networks typically run Low Power Devices
- Consist of one or more sensors, could be different type of sensors (or actuators)

Three main approaches for networking



Short-range multihop

- ZigBee
- WiFi low energy
- RFID



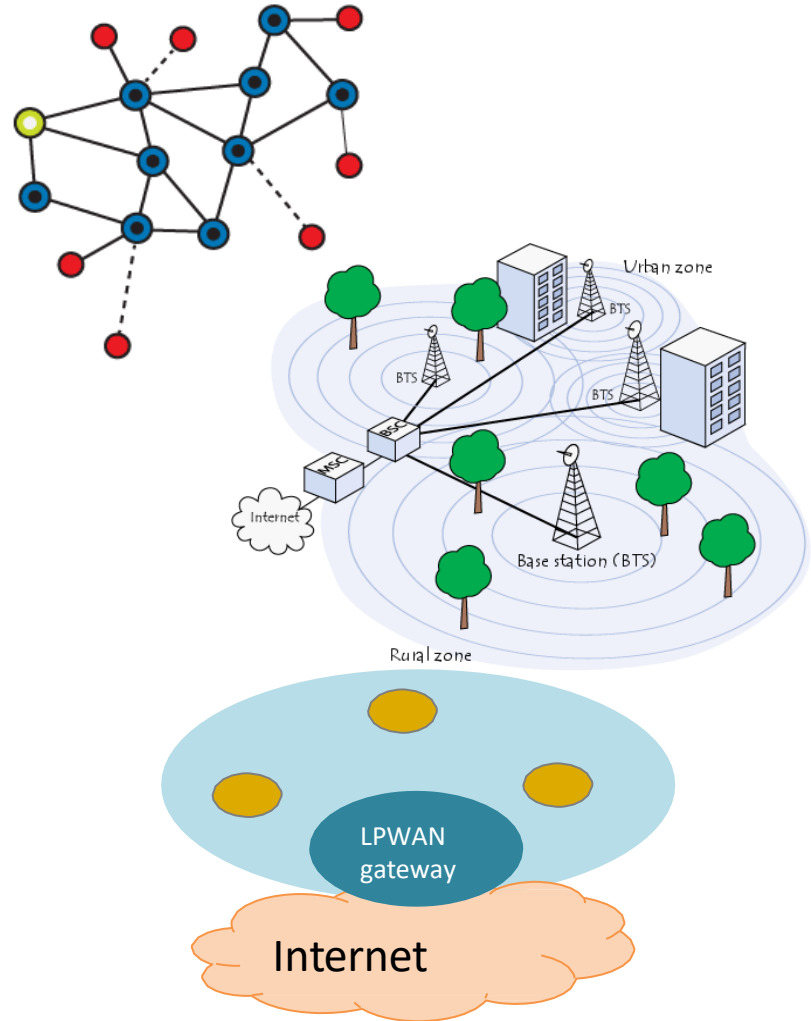
Cellular

- GSM
- LTE-A
- 5G

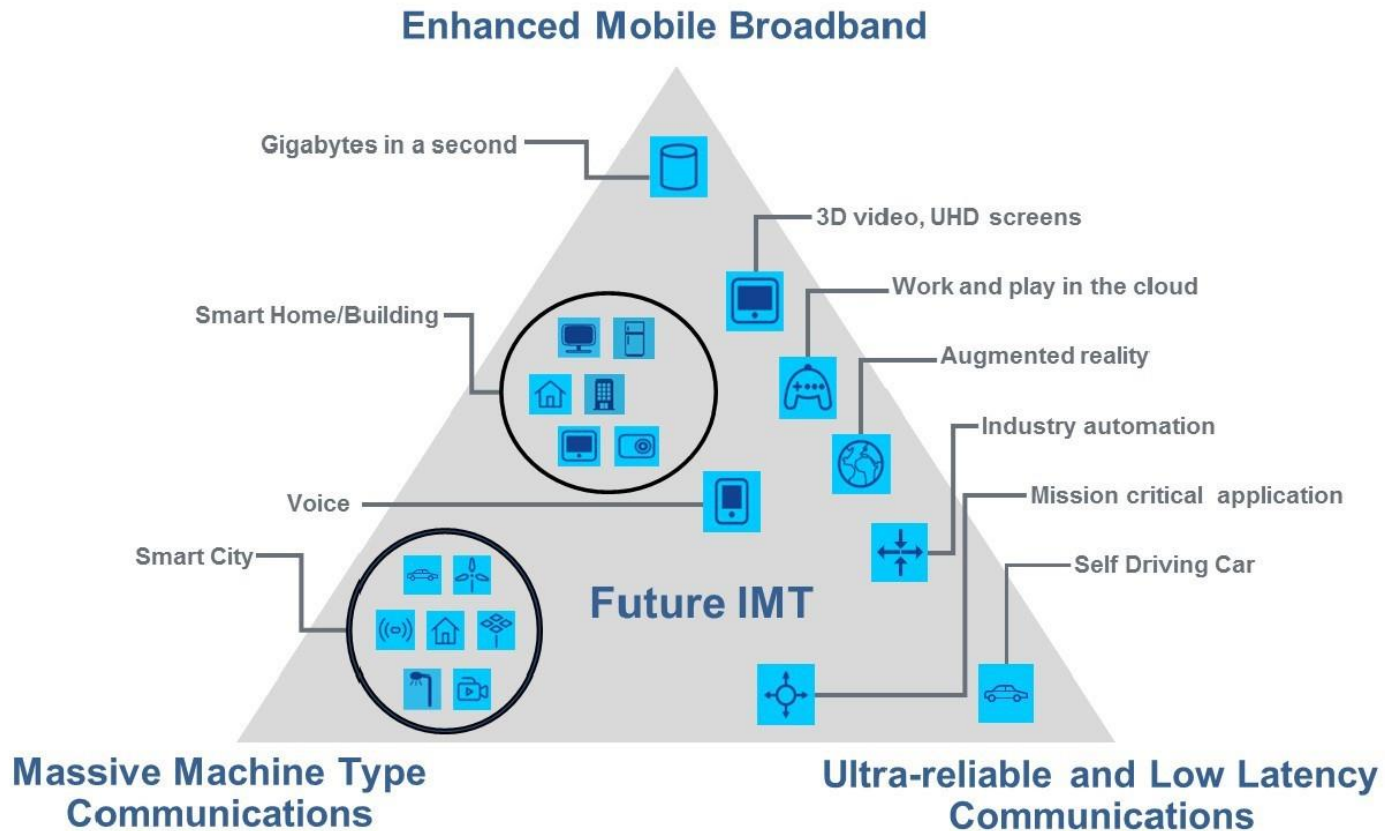


Low Power Wide Area Networks (LPWAN)

- SIGFOX
- NB-IoT
- LoRa



Use cases of 5G



Source: ITU-R M.2083-0, 2015

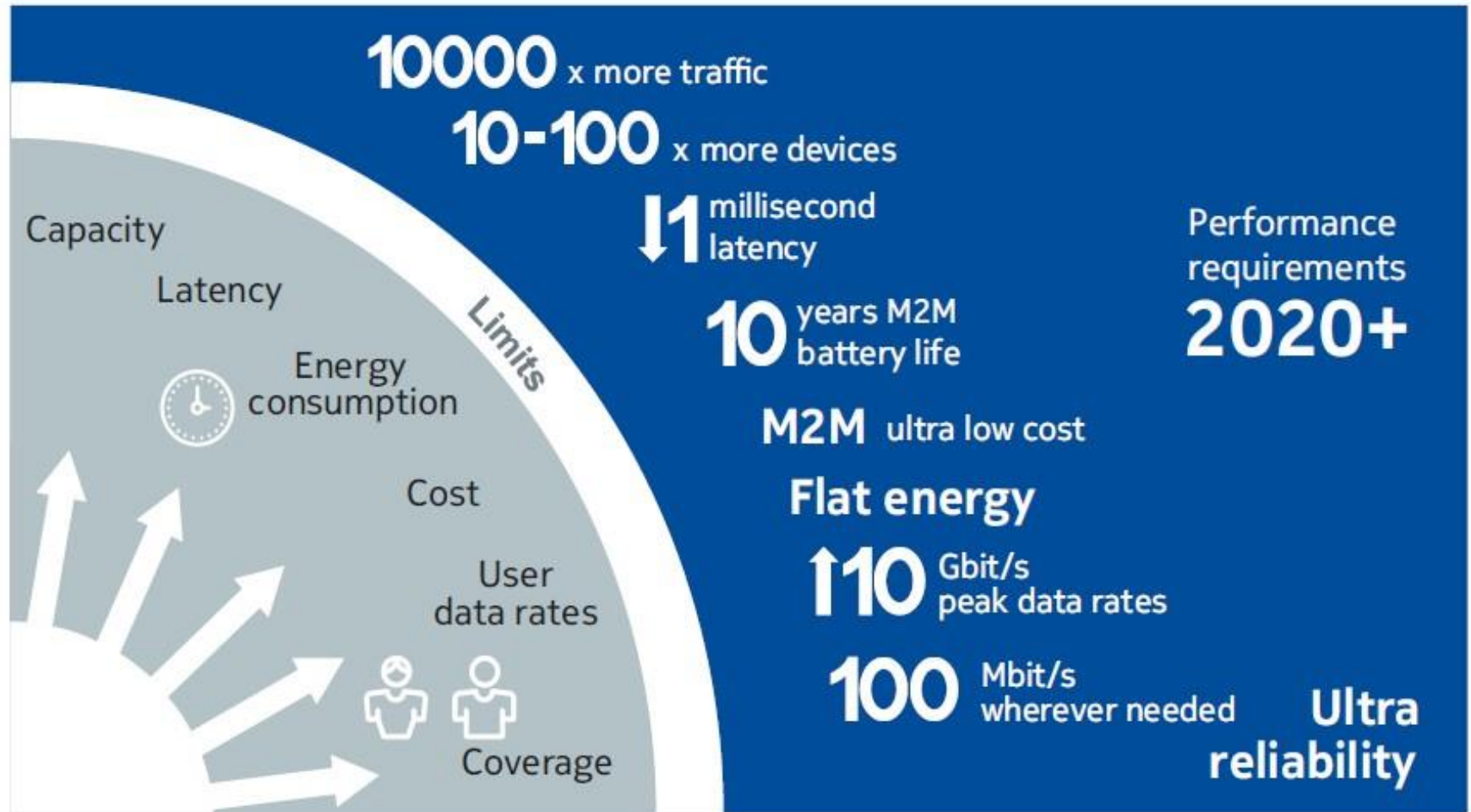
eMBB (enhanced mobile broadband), **xMBB** (extreme mobile broadband)

mMTC (massive machine type communications)

URLLC (Ultra reliable & low latency communications), **uMTC** (ultra reliable MTC)



5G requirements



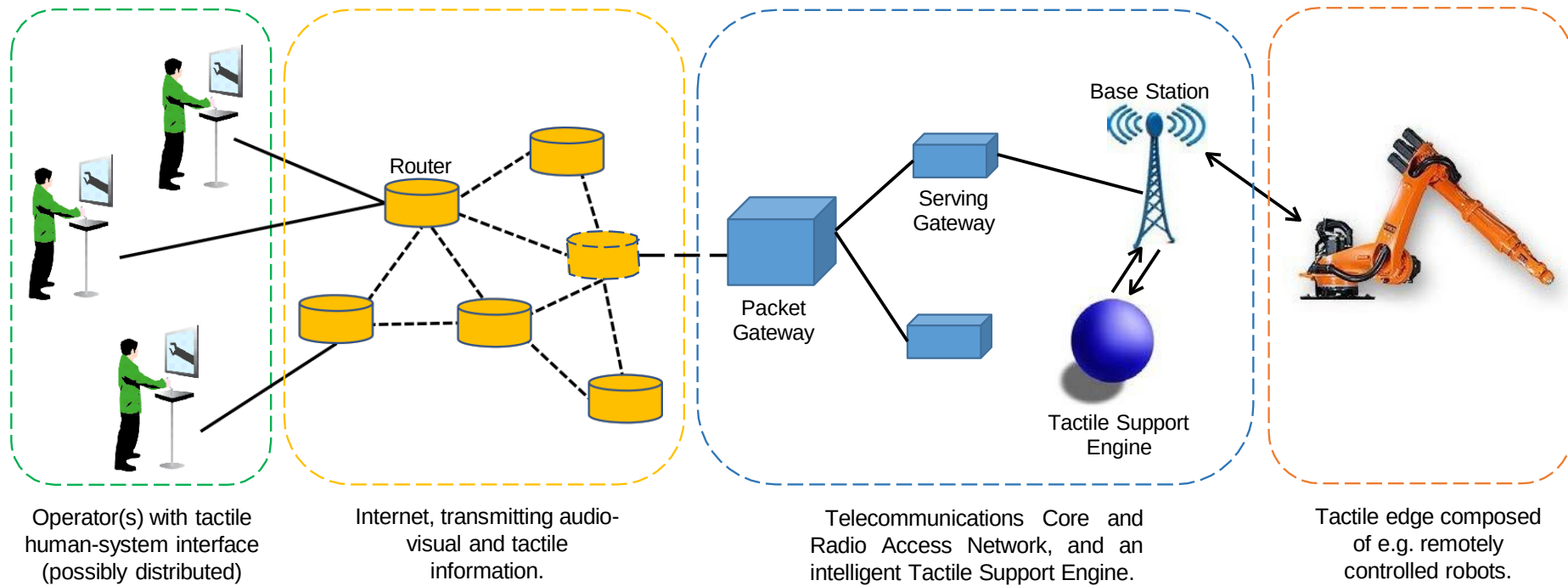
Source: *Understanding 5G*, Anritsu whitepaper, 2016





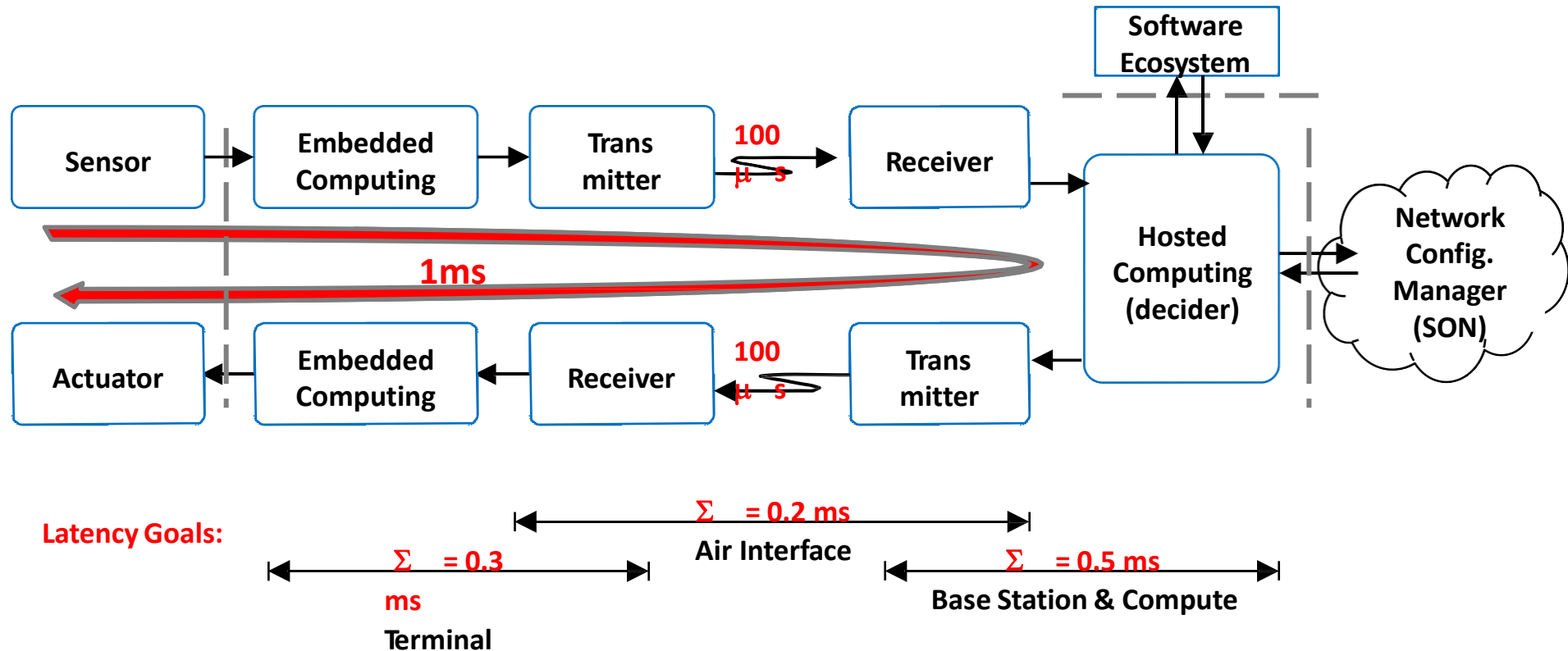
Source: <http://www.electronicshub.org>

Tactile Internet (2)



Source: Mischa Dohler, Keynote speech, NGMAST 2015

The goal of 1ms latency



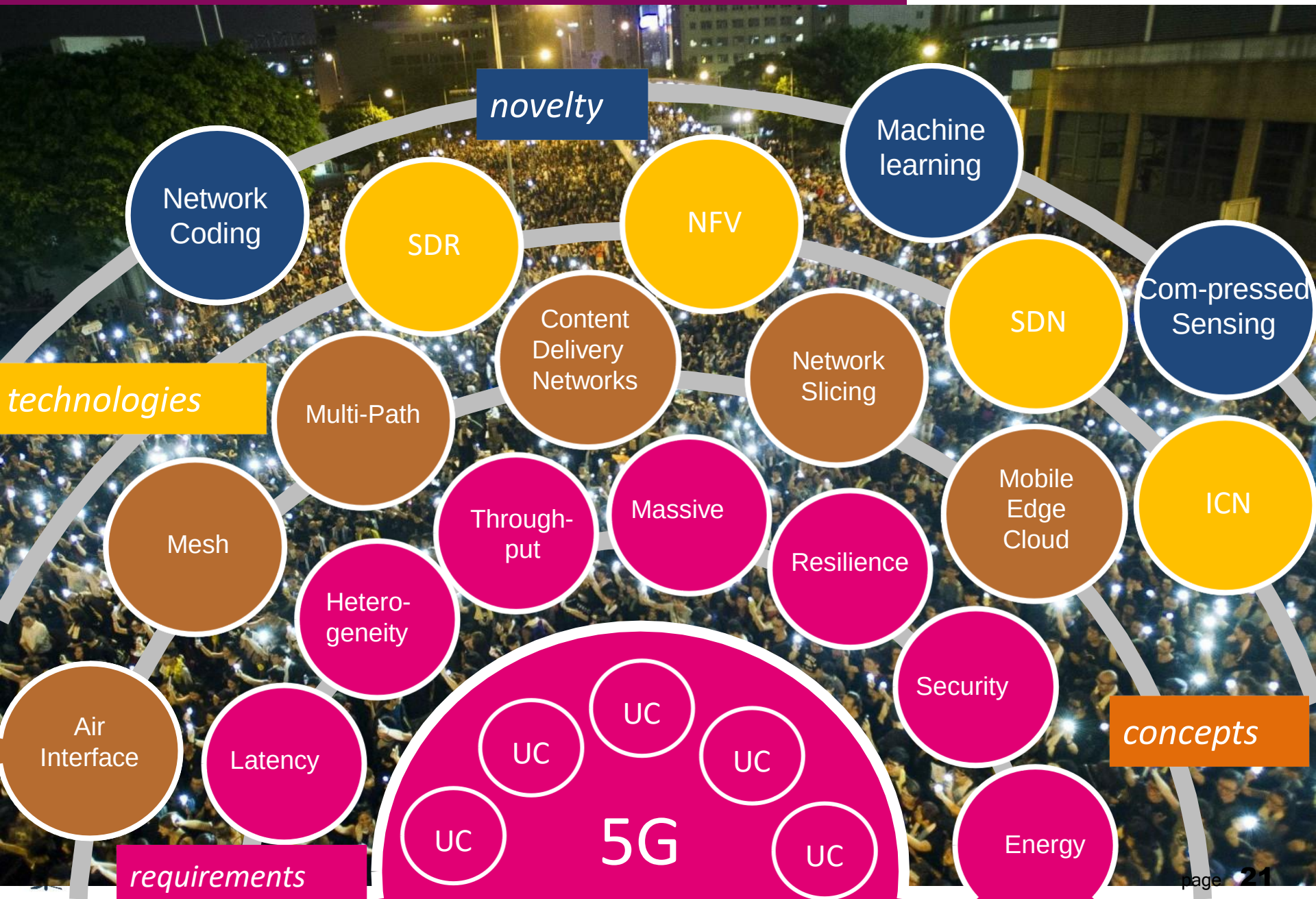
Precision Farming/Agriculture



Vehicle-to-everything (V2X)



The 5G world



5G / IoT challenges (summary)

- **Heterogeneity – Interoperability**
- **Scalability – Big Data**
- **Identification and Addressing**
- **Security – Privacy – Trust**
- **Miniaturization of devices**
- **Energy efficiency – Greening of IoT**
- **Standardization**



Thank you!

